

RAHW2

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Question 1

(i)

$$f(x, y) = \begin{bmatrix} y \sin(x^2) \\ e^{xy} \end{bmatrix} \quad (1)$$

$$(Df)_{xy} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} y \cos(x^2) 2x & \sin(x^2) \\ ye^{xy} & xe^{xy} \end{bmatrix} \quad (2)$$

(ii) & (iii) A is an inverse image of a set under a continuous map. A is defined $\forall \det(Df)_{xy} \neq 0$. We calculate this explicitly below in 4 but it can be seen by inspection. So we get the condition $(y \cos(x^2) 2x^2)(e^{xy}) - (ye^{xy})(\sin(x^2)) \neq 0$. Thus we have A defined $\forall y \neq 0$ and $2x^2 \neq \tan(x^2)$. This is open as it is a preimage of a continuous function is open. Open sets contain all its neighbourhoods. It doesn't contain its limit points e.g. $(0, 0)$ and thus is not closed.

If we show that the derivative of the inverse is the same as the inverse of the derivative then we can use that result.

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be continuously differentiable. Let $\mu(f(x, y)) = (x, y)$
 $D\mu(f(x, y))_{x_0, y_0} = D(x, y)_{x_0, y_0} = I$ by chain rule $\implies (D\mu|_{f(x, y)})(Df)_{x, y} = I$
 $(Df) = (Df)_{x, y}^{-1}$ And we know $(Df)_{x_0, y_0}^{-1}$ is given by

$$\left[(Df)_{xy} \right] = \frac{1}{\det(Df)_{xy}} \text{Adj}(Df) \quad (3)$$

$$= \left(\frac{1}{(y \cos(x^2) 2x^2)(e^{xy}) - (ye^{xy})(\sin(x^2))} \right) \begin{bmatrix} xe^{xy} & -\sin(x^2) \\ -ye^{xy} & y \cos(x^2) 2x \end{bmatrix} \quad (4)$$

$$= \frac{1}{ye^{xy}(\cos(x^2) 2x^2 - \sin(x^2))} \begin{bmatrix} xe^{xy} & -\sin(x^2) \\ -ye^{xy} & y \cos(x^2) 2x \end{bmatrix} \quad (5)$$

$$= \frac{1}{\cos(x^2) 2x^2 - \sin(x^2)} \begin{bmatrix} \frac{x}{y} & -\frac{\sin(x^2)}{ye^{xy}} \\ -1 & \frac{\cos(x^2) 2x}{e^{xy}} \end{bmatrix} \quad (6)$$

$$(Df)_{x_0, y_0}^{-1} = \frac{1}{\cos(x_0^2) 2x_0^2 - \sin(x_0^2)} \begin{bmatrix} \frac{x_0}{y_0} & -\frac{\sin(x_0^2)}{y_0 e^{x_0 y_0}} \\ -1 & \frac{\cos(x_0^2) 2x_0}{e^{x_0 y_0}} \end{bmatrix} \quad (7)$$

(iv) We have $\phi: X \rightarrow \mathbb{R}^m$ continuously differentiable on open X with $\det(D\phi)_{\vec{x}} \neq 0, \forall \vec{x} \in X$.

Since $\det(D\phi)_{\vec{x}} \neq 0 \implies \phi$ is invertible and it is continuously differentiable with local inverse $\mu(\phi(\vec{x})) = \vec{x}$. Thus the inverse function theorem applies.

Let $\vec{x}_i \in X$ and U_i be a neighbourhood around x_i . Define $X \equiv \bigcup_i U_i$. Let $Y \equiv \bigcup_i Y_i$ for some $Y \subset \mathbb{R}^m$.

Injectivity

$\phi|_{U_i} U_i \rightarrow Y_i$. Let x_1 and x_2 be elements of U_i . If $\phi(x_1) = \phi(x_2)$, inverting gives $\mu(\phi(\vec{x}_1)) = \mu(\phi(\vec{x}_2))$, and $\vec{x}_1 = \vec{x}_2$. Then $\phi|_{U_i} U_i$ is injective. So we have $\phi(U_i) \subseteq Y$.

Surjectivity

$\forall \vec{y} \in Y \exists \vec{x} \in X$ such that $\phi(\vec{x}) \in Y$. Thus ϕ is surjective. So we have $Y \subseteq \phi(U_i)$.

Since $\phi(U) \subseteq Y$ and $Y \subseteq \phi(U)$. We have $\phi(U) \equiv Y$. So $\phi: X \rightarrow Y$ is bijective.

Since ϕ is bijective and a continuously differentiable it is a diffeomorphism. We know diffeomorphisms preserve openness. Thus $Y \equiv \phi(U_i)$ is open. We have $\bigcup_i \phi(U_i) = \phi(X)$ which is open as it is a union of open sets.

Question 2 (i)

let $S = \{(x, y) \in \mathbb{R}^2 | y^2 = x^3 + x^2\}$.

IFT requires the rank of the jacobian is at least equal to the number of rows in $(DF)_{\vec{p}}$. Let $f(x, y) = 0 = x^3 + x^2 - y^2$.

Then we have $\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = [3x^2 + 2x, -2y]$

$\text{Rank}(Df) \geq 1$. The only case where this isn't satisfied is if $\text{Rank}(Df) = 0$. In which, both entries must be zero. In that case we have $y = 0$ and $3x^2 + 2x = 0 \implies x(3x + 2) = 0$. So either $x = 0$ or $x = -\frac{2}{3}$. If we plug these back into S we see $x = -\frac{2}{3}$ doesn't satisfy it as $0 \neq (-\frac{2}{3})^3 + (-\frac{2}{3})^2$. Thus the only point where the IFT doesn't apply is at $(x, y) = (0, 0)$. So S_0 is given by $S_0 = S \setminus \{0\}$. Thus S_0 isn't closed in S as it doesn't contain $(0, 0)$. S_0 is open in S as it contains all open balls (intervals) in S .

(ii)

We have a graph of the function below:

We see a curve which intersects itself near the origin. This gives us four branches.

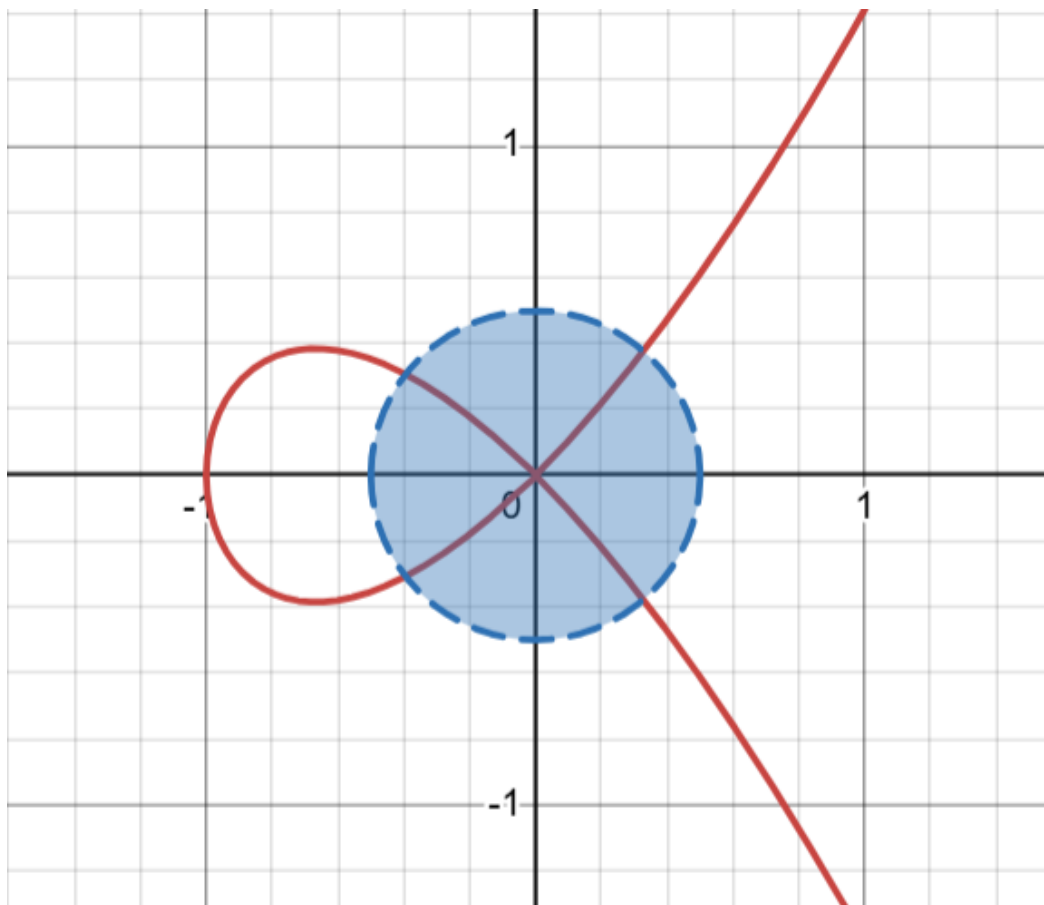


Figure 1:

In order for $V \cap S$ to be homeomorphic it must be a bijective and continuous function between spaces with a continuous inverse function. We see that since there are four branches an interval (a, b) can't be a bijection. Thus there doesn't exist an (a, b) such that $V \cap S$ is homeomorphic to (a, b) .